



Large Language Models

From ZERO To HERO

NLP

NLP is a subfield of AI that focuses on developing technologies that are capable of understanding, interpreting, and generating human language.

LANGUAGE MODEL

A Language Model (LM) is any model that learns to predict the next word (or token) in a sequence – based on what came before.

LANGUAGE MODEL

CORE OBJECTIVE

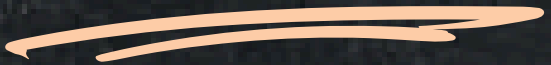
$P(\text{next_token} \mid \text{previous_tokens})$

TYPES OF LANGUAGE MODELS

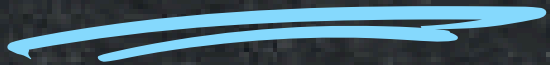
STATS ERA



NEURAL ERA



TRANSFORMERS



TYPES OF LANGUAGE MODELS

STATS ERA

N-grams, Markov
Chains

NEURAL ERA

RNN, LSTM

TRANSFORMERS

ENCODER-DECODER,
DECODER-ONLY

LARGE LANGUAGE MODELS

A Large Language Model (LLM) is a Transformer-based Language Model that's trained at massive scale. So while the core idea is still "predict the next token," the scale and architecture make all the difference.

TYPES OF LLMS

ENCODER-DECODER

T5

ENCODER ONLY

BERT

DECODER ONLY

GPT

TOKENIZATION & TOKEN ID

Tokenization is the process of breaking down text into smaller units called tokens. These tokens can be words, subwords, characters, or even sentences, depending on the level of granularity required for a specific application.

TOKENIZATION & TOKEN ID

There's one more step involved after breaking down the text into tokens, which is converting these tokens into numerical representations (like token IDs) that the model can process.

TOKENIZER

Tokenizer is a tool or algorithm that performs tokenization. Every language model has its own tokenizer that is specifically designed to work with that model. That's why when you use different models, you often need to use their respective tokenizers to ensure compatibility.

EMBEDDINGS

Embeddings are vector representations of words that attempt to capture its meaning.

EMBEDDINGS

TYPES OF EMBEDDINGS

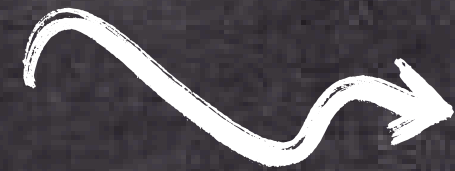
STATIC

Same and constant for all the processing.

CONTEXTUAL

It changes as per the sentence and context

EXAMPLE



THE WHITE CAT SAT ON THE



TOKENIZER

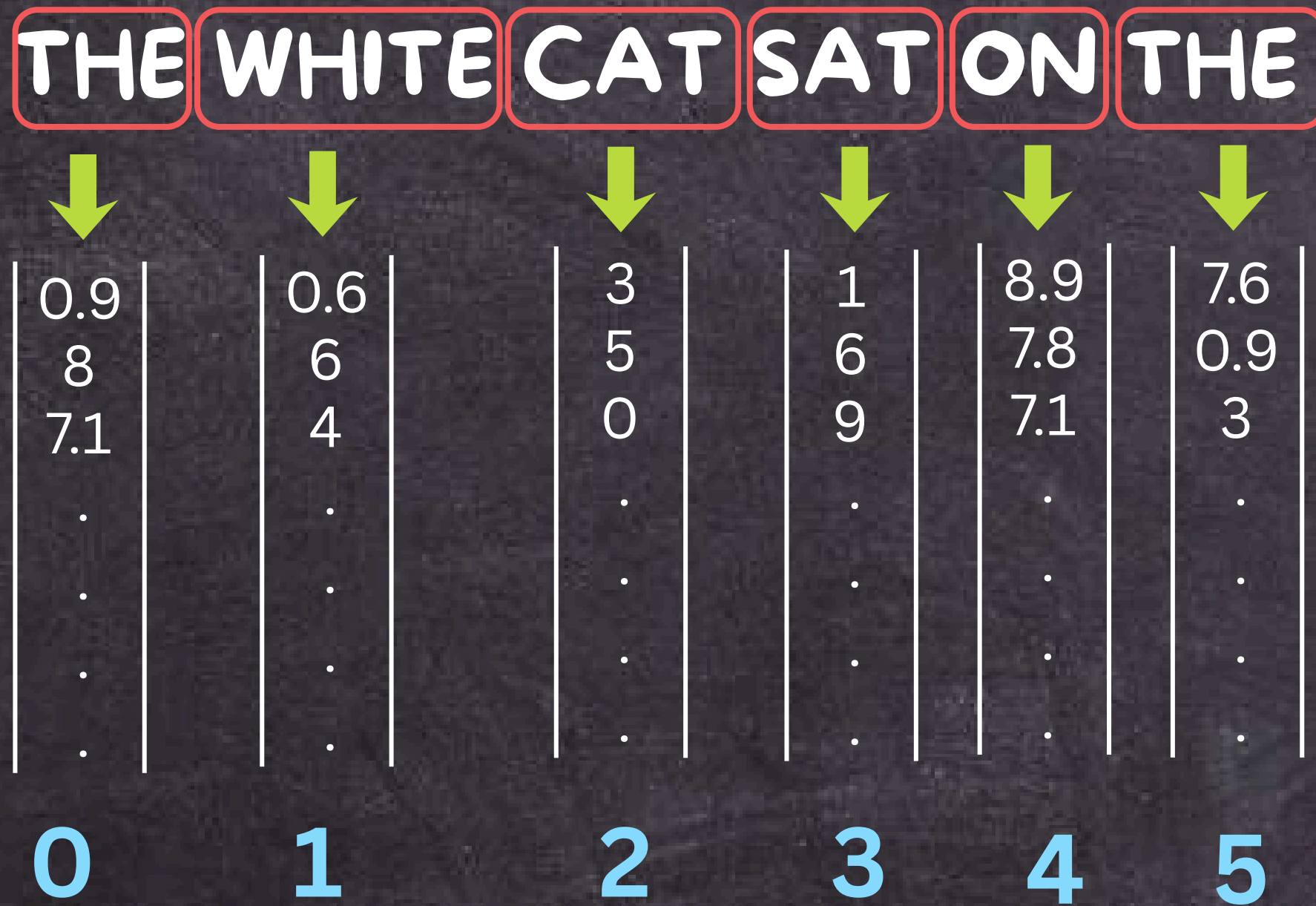
THE WHITE CAT SAT ON THE

STATIC EMBEDDING

THE WHITE CAT SAT ON THE

↓	↓	↓	↓	↓	↓
0.9	0.6	3	1	8.9	7.6
8	6	5	6	7.8	0.9
7.1	4	0	9	7.1	3
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.

POSITIONAL ENCODING



GOAL

THE WHITE CAT SAT ON THE

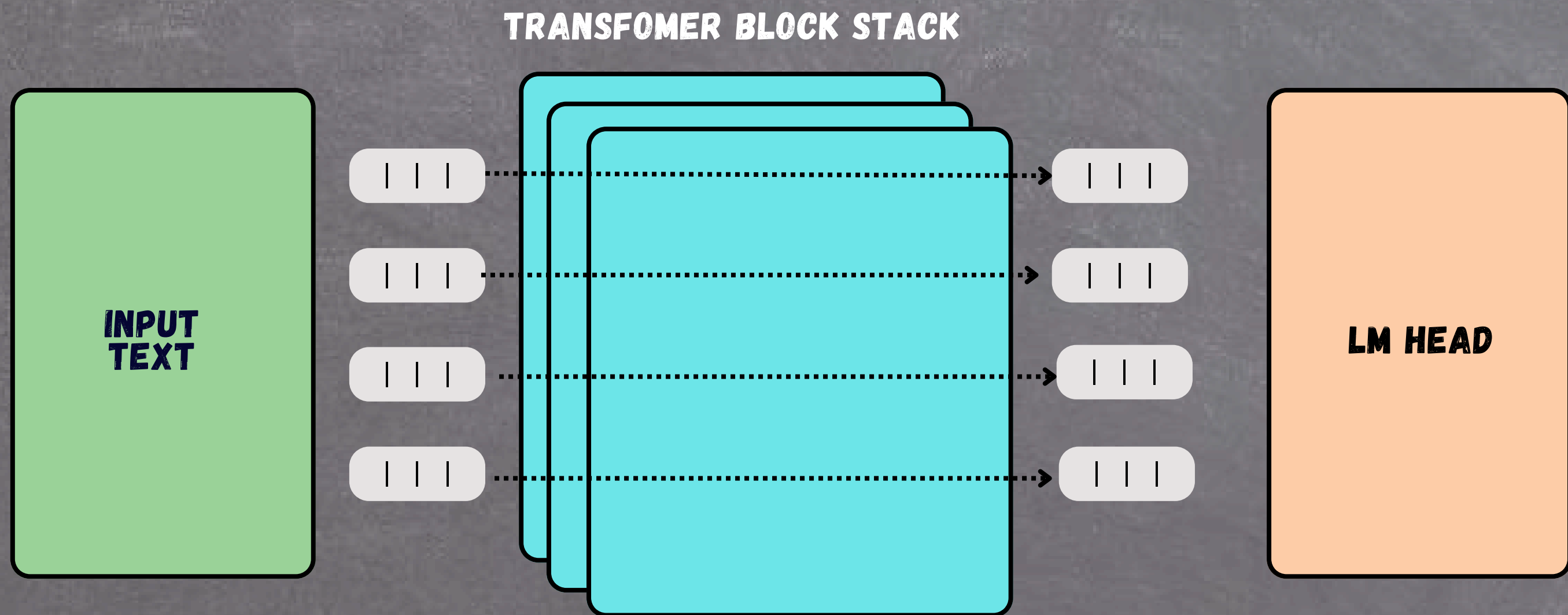


E1 E2 E3 E4 E5 E6

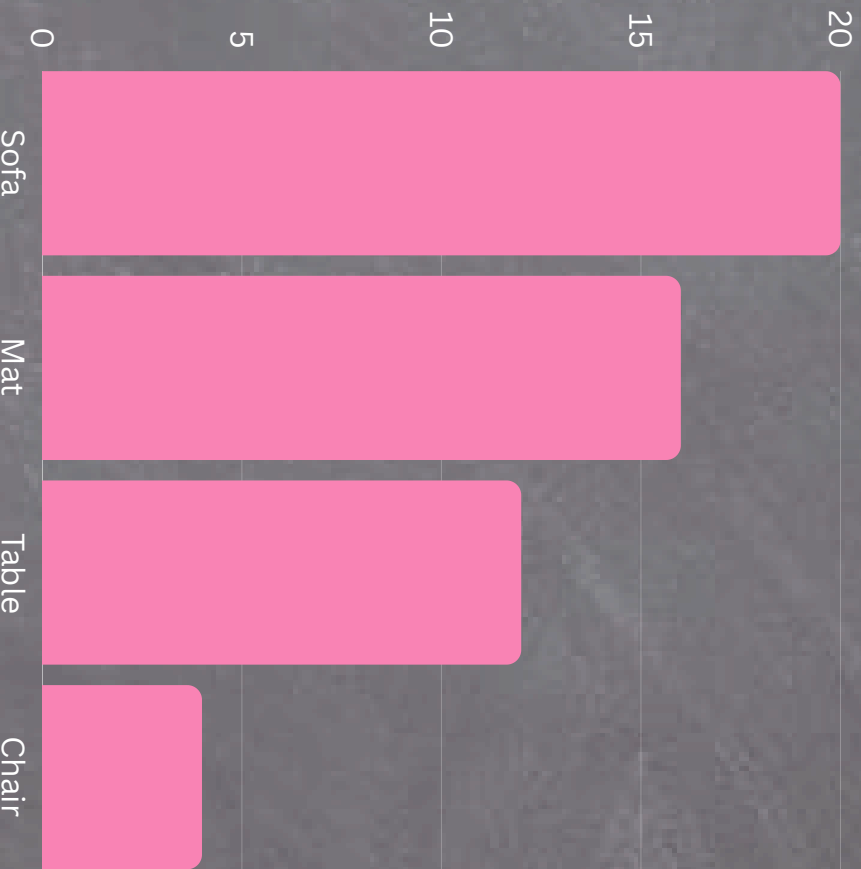
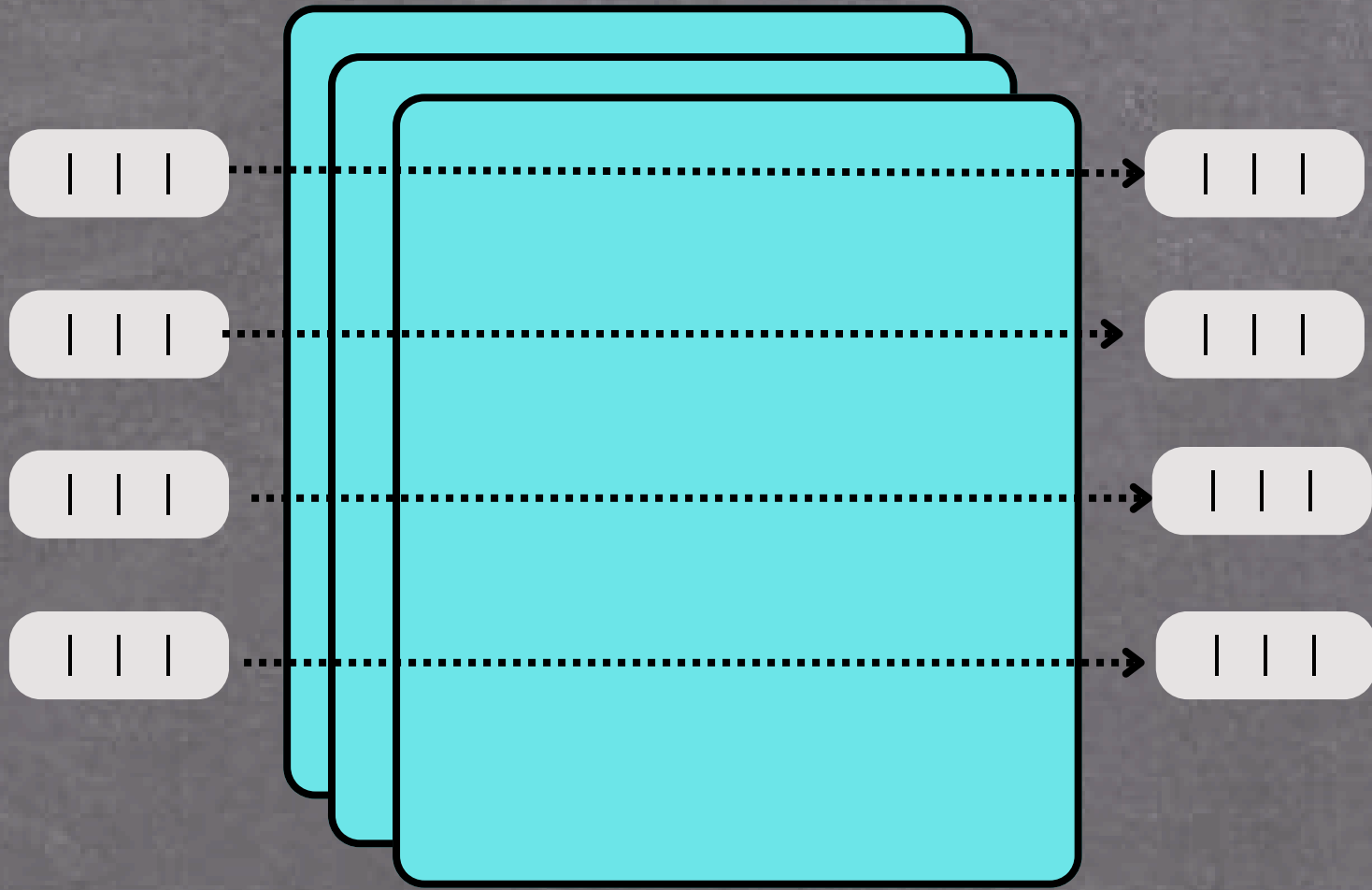


E1' E2' E3' E4' E5' E6'

CONTEXTUAL EMBEDDINGS



TRANSFOMER BLOCK STACK



SELF ATTENTION



QUERY

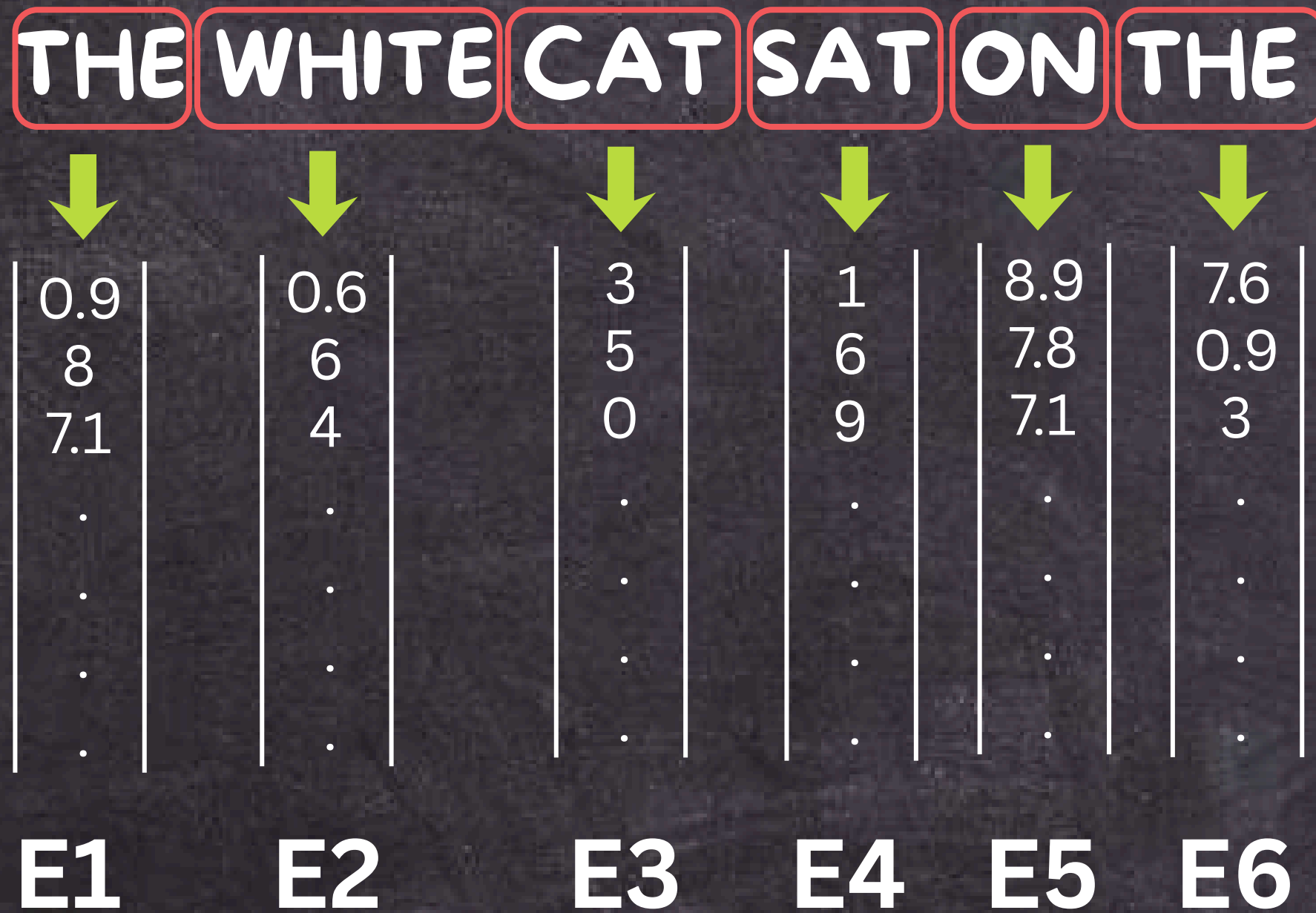


KEY



VALUE

SELF ATTENTION



QUERY (Q)

CAT



3
5
0
.
.
.
.

E3

W(Q)

3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1

Q3

=

0.3
3.2
-1
.
.
.
.

QUERY (Q)

E1

E2

E3

E4

E5

E6



Q1

Q2

Q3

Q4

Q5

Q6

KEY (K)

WHITE



3
5
0
.
.
.

E3

W(K)

3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1

=

K3

0.3
3.2
-1
.
.
.

KEY (K)

E1

E2

E3

E4

E5

E6



K1

k2

K3

K4

K5

K6

RELEVANCY SCORE

	Q1	Q2	Q3	Q4	Q5	Q6
K1	0.7	12	91	1	8	0.7
K2	0.6	23	98	23	0.6	0.6
K3	0.5	0.9	21	0.8	8	0.5
K4	-21	11	-21	-21	13	2
K5	-42	21	0.9	5	7	-42
K6	0.1	10	0	1	1	2

MASKING

	Q1	Q2	Q3	Q4	Q5	Q6
K1	0.7	12	91	1	8	0.7
K2	$-\infty$	23	98	23	0.6	0.6
K3	$-\infty$	$-\infty$	$-\infty$	0.8	8	0.5
K4	$-\infty$	$-\infty$	$-\infty$	$-\infty$	13	2
K5	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	-42
K6	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	2

SOFTMAX

	Q1	Q2	Q3	Q4	Q5	Q6
K1	0.7	0.2	0.1	0.01	0.1	0.1
K2	0	0.8	0.8	0.4	0.4	0.3
K3	0	0	0.1	0.5	0.2	0.3
K4	0	0	0	0.1	0.2	0.1
K5	0	0	0	0	0.1	0.1
K6	0	0	0	0	0	0.1

VALUE (V)

CAT



3
5
0
.
.
.
.

E3

W(V)

3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1
3.8, 1.9, -08, 0.6, -1

=

V3

0.3
3.2
-1
.
.
.
.

DELTA CONTEXT

CAT

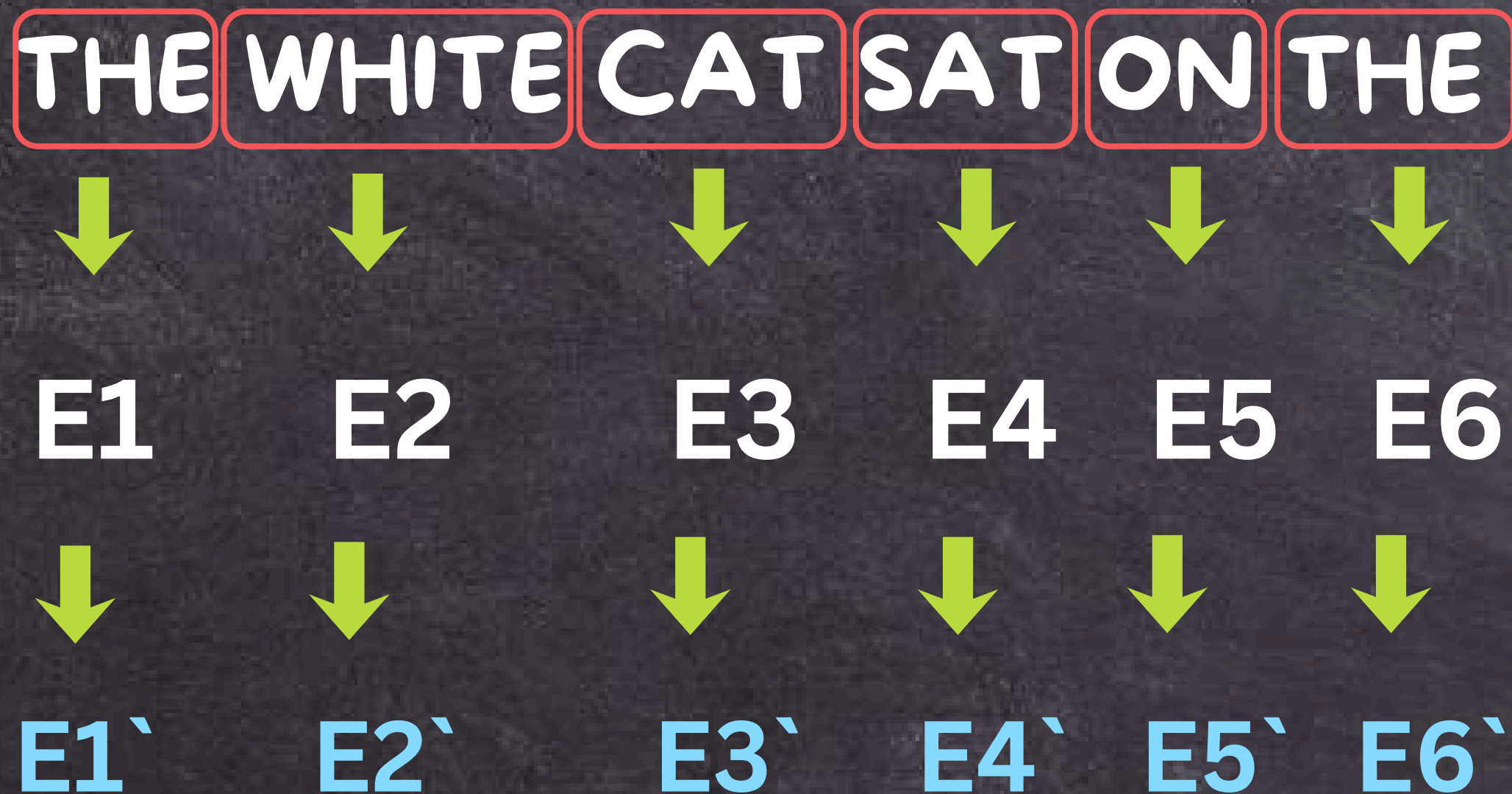
$$\Delta E3 = V1*0.1 + V2*0.9 + \dots$$

In short, 10% of “The” and 90% of “White” we are having a DELTA CONTEXT

FINAL CONTEXTUAL EMBEDDING

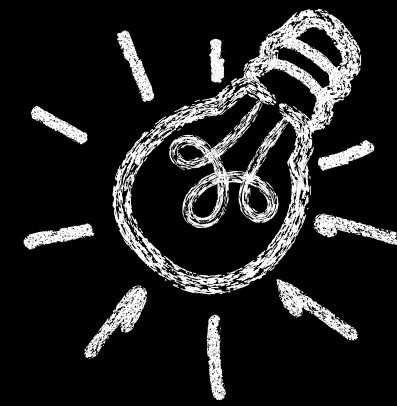
$$E3' = E3 + \Delta E3$$

FINAL CONTEXTUAL EMBEDDING



CONTEXTUAL EMBEDDINGS

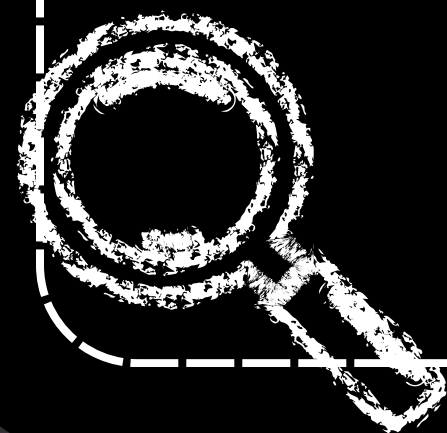
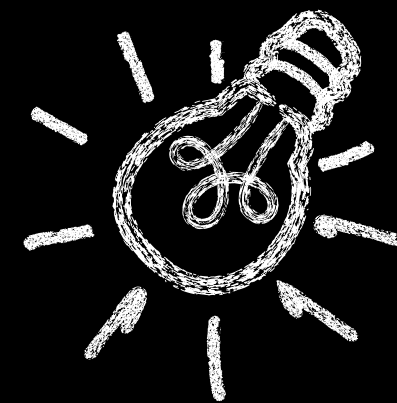
ENCODER-ONLY TRANSFORMER



Encoder-Only transformers are designed primarily for understanding and processing input sequences, making them ideal for tasks such as text classification, sentiment analysis, and named entity recognition.

By architecture, it is exactly same as the Decoder-Only transformer we discussed earlier, except that it does not have the mechanism to generate output sequences. Instead, it focuses on encoding the input data into rich contextual embeddings that can be used for various downstream tasks.

ENCODER-DECODER TRANSFORMER

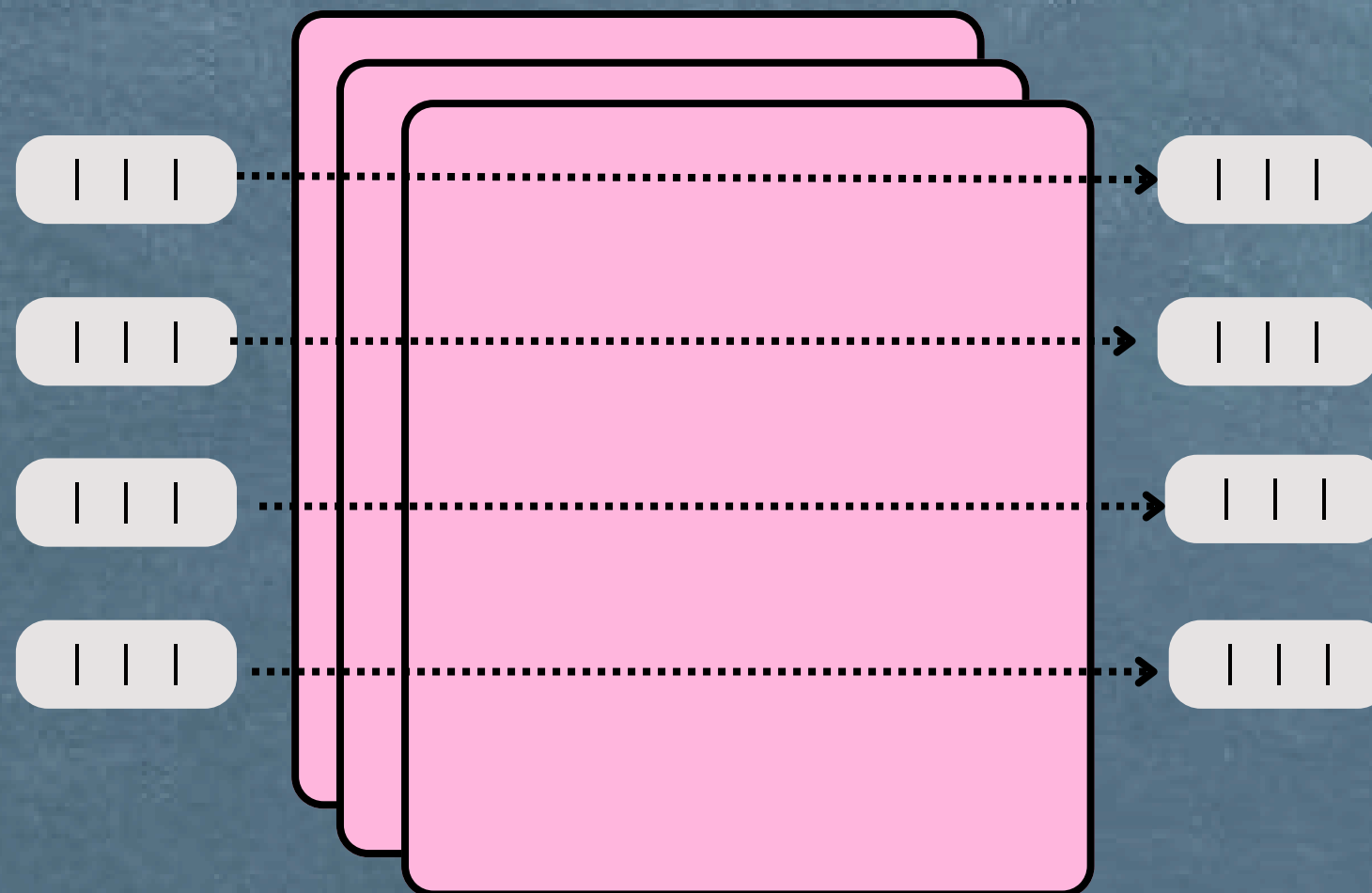


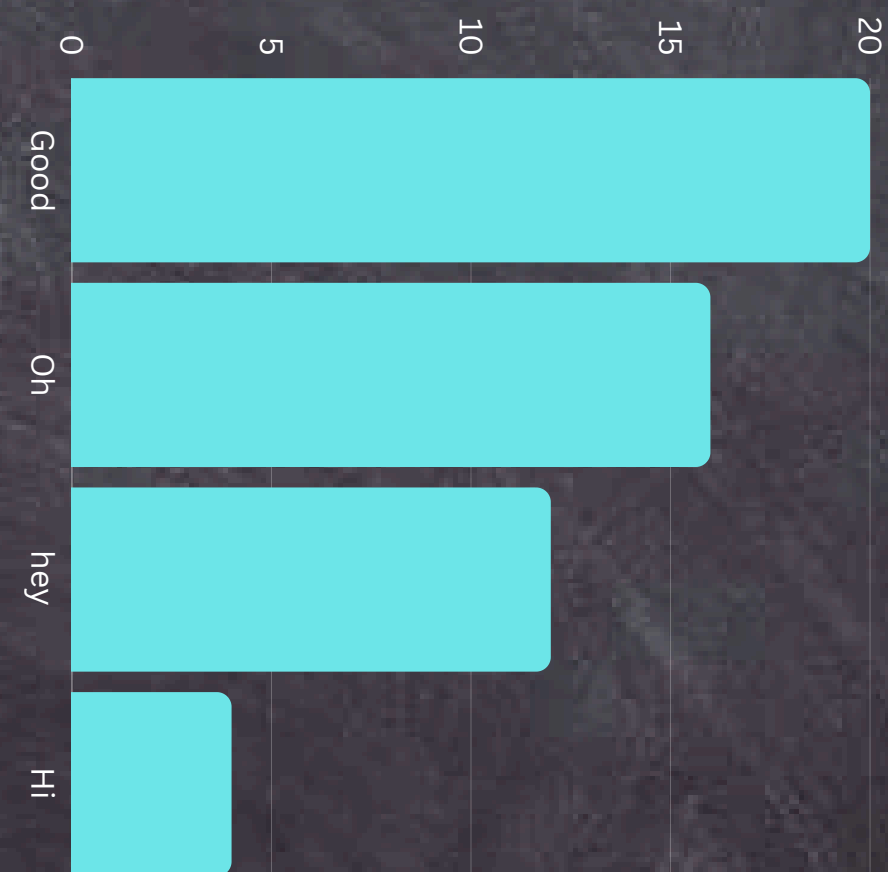
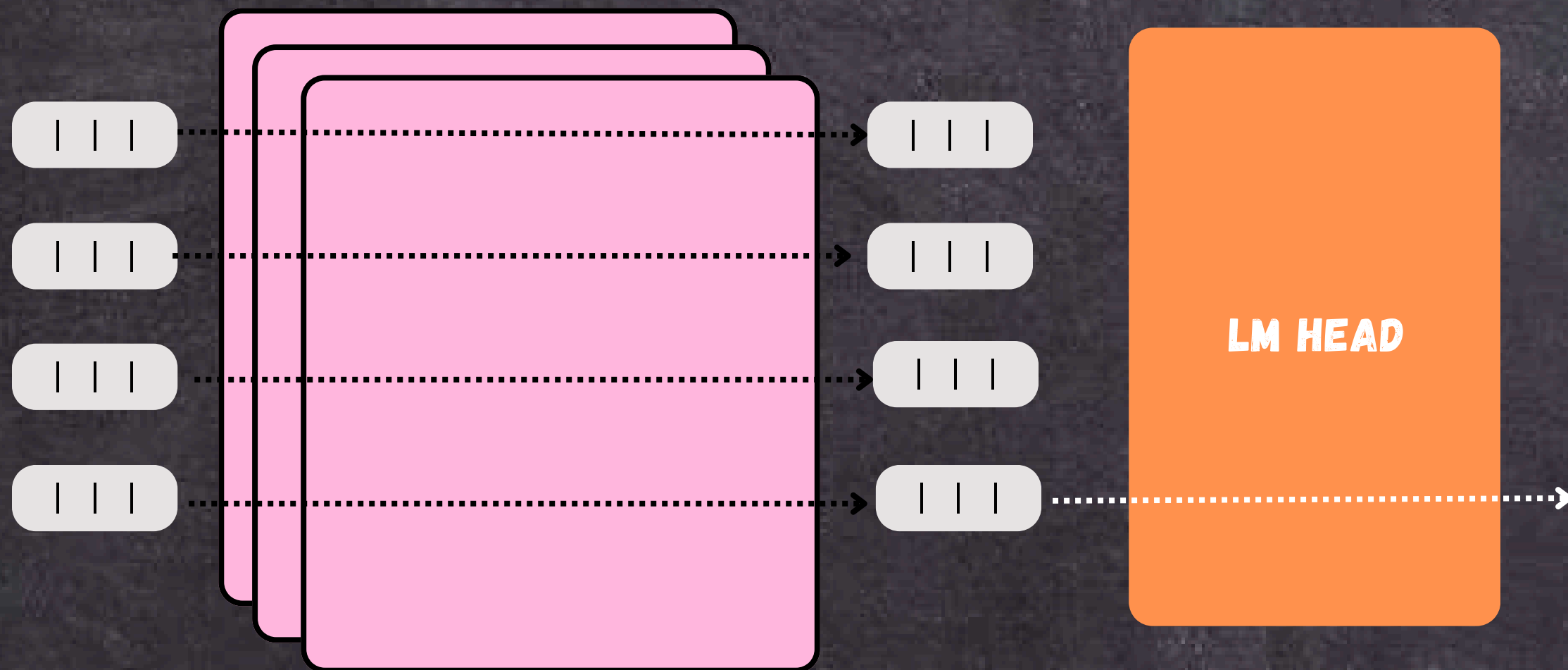


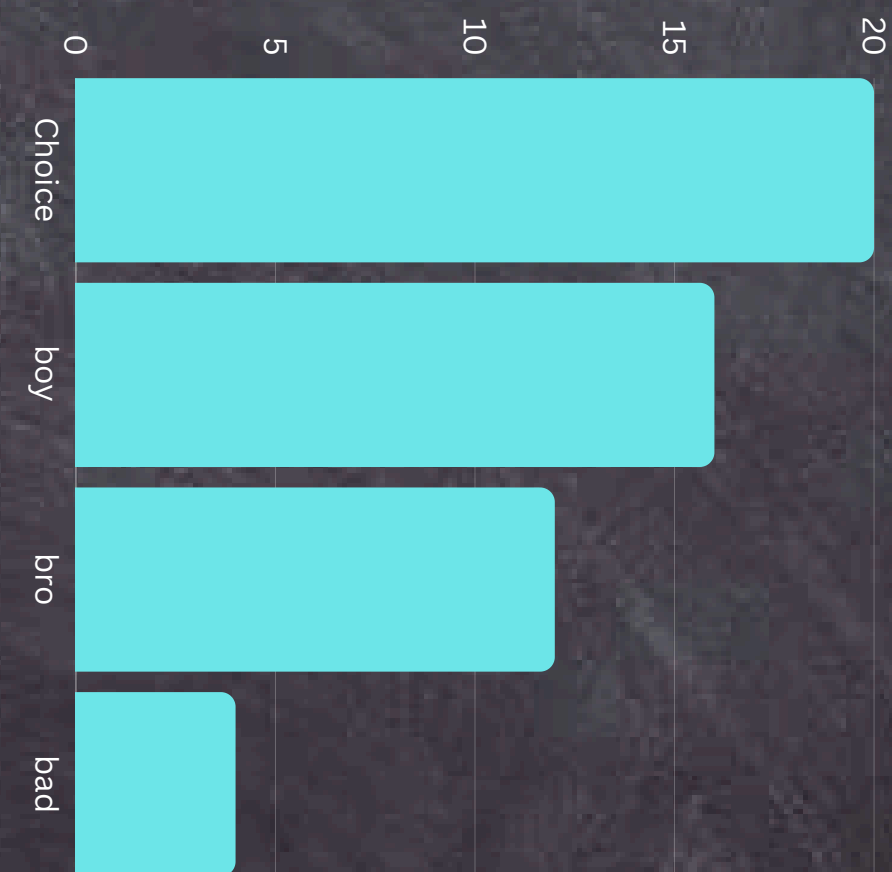
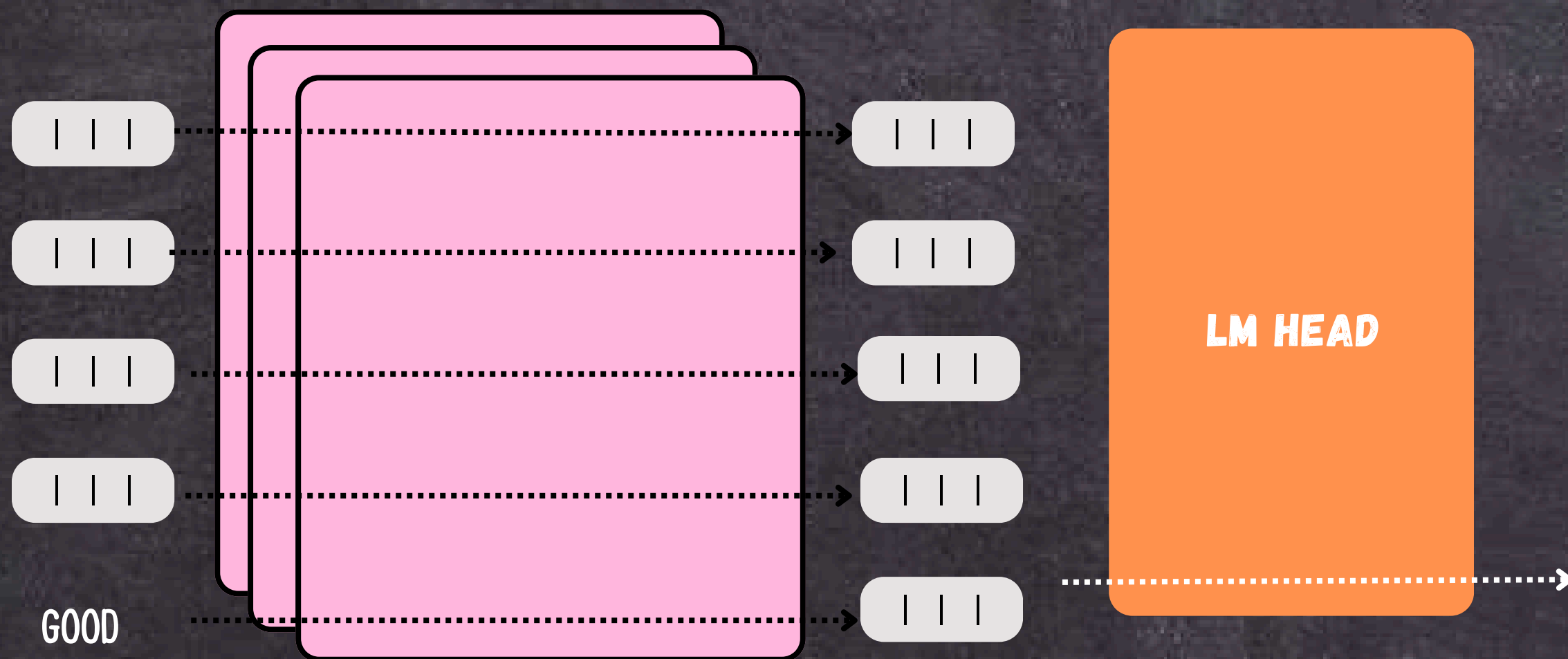
TOKENIZER

TRANSFOMER BLOCK STACK

LM HEAD







 @anaghak.s8750 · 15 hours ago
You're amazing! Your content has been a huge help in my Azure data journey. I'm excited to receive an offer and joined a new organization yesterday. Thank you so much for your support!

Reply

0 replies ^



 @AnshLambaJSR · 0 seconds ago
Wow, congratulations!

Reply



 @pavankumar12121 · 8 hours ago
I just wanted to let you know that I got the job — and it honestly wouldn't have been possible without the projects you helped me with. Thank you so much from the bottom of my heart. I'm really, really grateful!

Reply

2 replies ^



 @amoltare2843 · 3 hours ago
Same words ❤️ ..He is a saviour

Reply



 @ShafiqAhmed-h7t · 10 hours ago

I got placed in two companies waiting for more all kudos to your help and videos and thank you a ton 🙏🙏🙏 am grateful to you am learning a lot from you I have 10 years of experience and the knowledge I got from past two months from you is precious you are a gem and a jewel bro.

Reply

4 replies ^



 @prateekjha5162 · 15 hours ago

Bro, I do have 7 years of exp in DE already.. about to receive 2 offers because of you (more incoming offers and even service notice period which is 90 days for me !!! thank you so much! If you're in India sometime, please let the community know so that we can sit together at least once

Reply

0 replies ^



 @AnshLambaJSR · 0 seconds ago
Proud of my DATA FAM :)

 @mohithkariappak6148 · 7 hours ago
Hi Ansh,

Just wanted to take a moment to thank you. I have cracked 2 offers watching your videos. Your videos have had a great impact during my preparations for the interviews and learnt a lot of new things at the same time.

Kudos to your hard work. Will keep following you throughout my data engineer career.

Reply

0 replies ^



 @DEEPAKPARMAR-u5j2c · 7 hours ago

Hey Bro, Thanks for all practical knowledge. I am able to crack 2 offers from your contents. I hope you are providing all this content is high in quality

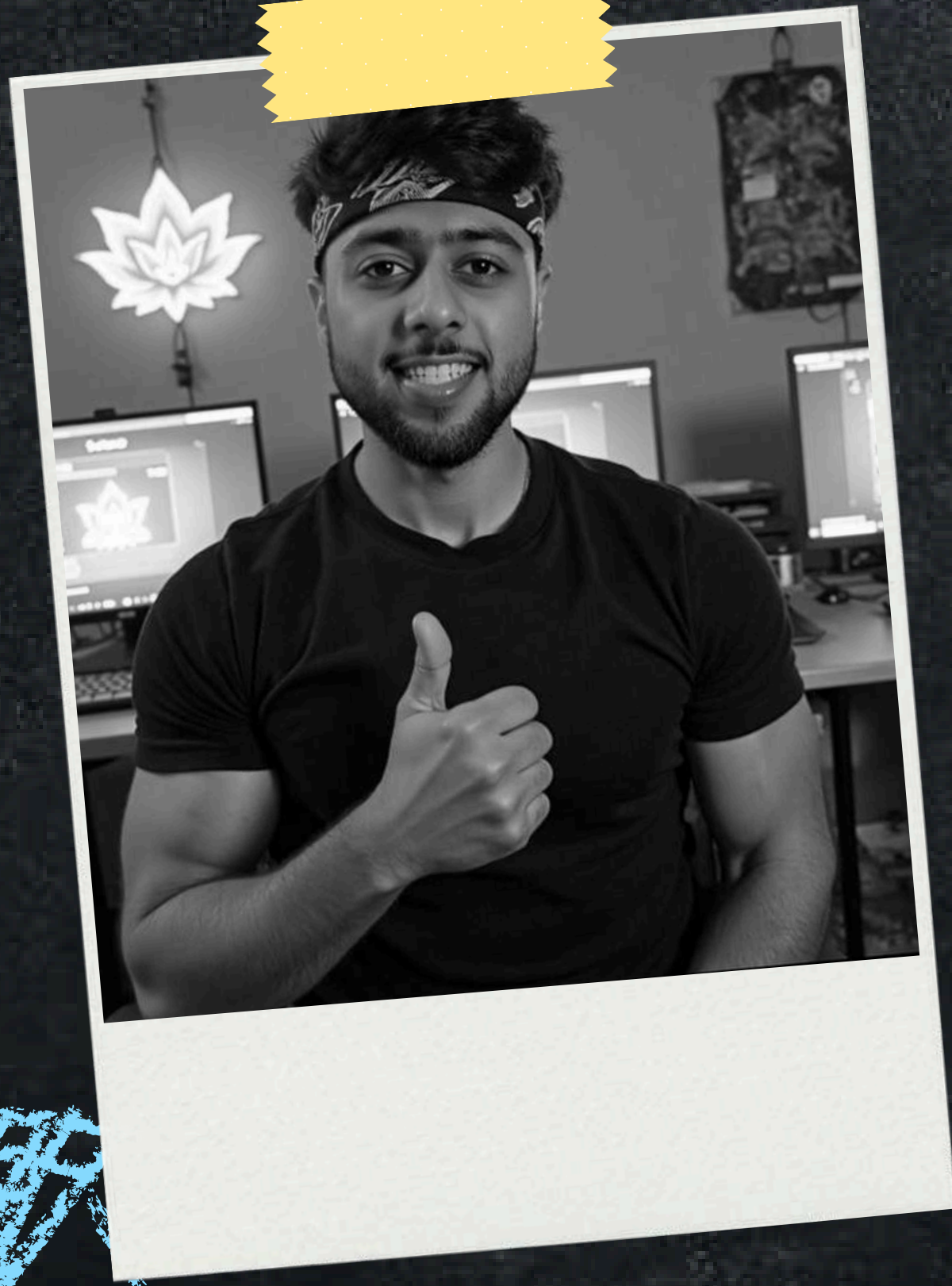
Reply

1 reply ^



 @miky7595 · 7 hours ago

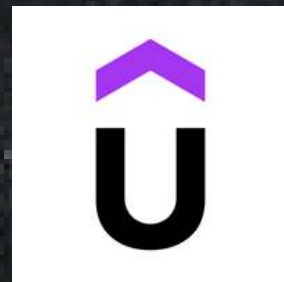
I got 5 offers after preparing from your videos. A big thank you to you bro from the bottom of my heart. I was under the confidence of cracking even a single round of any company 3-4 months back.. Happy birthday to you, shining



HANDLES



ANSH LAMBA



ANSH LAMBA JSR